

Elio Govea

Software Developer

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Profile

Software developer. Currently building aviation weather systems at [Vaisala](#). Former [ICPC](#) World Finalist

Experience

Senior Software Engineer [Vaisala](#)

Vantaa, Finland 02/2025 - present

- Contributing to [AviMet](#), an aviation weather information system used by airports and air-traffic services

Senior Developer [F-Secure](#)

Helsinki, Finland 05/2023 - 02/2025

- Designed solutions to reduce Content Delivery Network (CDN) costs by 80% with minimal impact on end-users
 - Analyzed CDN usage and identified key optimization areas
 - Designed a custom caching strategy to significantly decrease the volume of data transferred via CDN
 - Ensured that changes were backend-focused to avoid client-side updates and lengthy deployment negotiations

Developer [F-Secure](#)

Helsinki, Finland 12/2021 - 04/2023

- Built an SDK providing security features for home routers
 - Developed most features using C++
 - Supported multiple architectures
 - x86 — ARM — MIPS
 - Supported filtering, parsing and generating traffic for multiple network protocols
 - IPv4 — IPv6 — DHCP — UDP — TCP — DNS — HTTP — HTTPS
 - Used multiple Linux interfaces and utilities
 - netlink — netfilter — iptables — nfqueue — nflog — conntrack
 - Integrated functionalities to cloud services
 - Collaborated closely with cross-functional teams to gather requirements, iterate on designs, and address specific needs of diverse projects.
- Developed applications that secure devices connected to home routers
 - Implemented features to prevent clients from accessing malicious content on the internet
 - Integrated to existing back-end services
- Developed parental-control features for home devices
 - Implemented features to force clients to resolve safe versions of major browser engines
 - Integrated to existing back-end services
- Designed and integrated features to collect traffic statistics
 - Integrated to back-end services to detect traffic anomalies
 - Minimized effects on networking performance
 - Used multiple Linux interfaces: nfqueue — nflog — conntrack
- Implemented features to recognize/fingerprint devices
 - Implemented features to filter multi-cast traffic

- Used several Linux interfaces: nflog — BPF
- Minimized effects on networking performance
- Integrated to multiple back-end services
- Implemented recovery mechanisms for OS- and cloud-service-level errors
- Developed SDK integration to reference platforms
 - Simplified products demos
 - Simplified future integration
- Designed and optimized the build system of multiple cross platform C/C++ applications
 - Supported multiple toolchains
 - Supported multiple architectures
 - Provided fast and reproducible builds
 - Implemented with CMake and GNU Make
- Designed and implemented continuous integration pipelines
 - Enforced code quality
 - Code coverage — Static analysis — Dynamic analysis
 - Provided builds for multiple architectures
 - x86 — ARM — MIPS
 - Provided builds for reference platforms
 - Implemented using Jenkins and AWS
- Designed and implemented testing tools
 - Python — Robot Framework

Software Developer *F-Secure*

Poznań, Poland 09/2020 - 11/2021

- Contributed to a core C++ SDK providing security features for home routers
- Optimized a notifications system for security events
- Developed embedded Linux clients to consume cloud services
- Added endpoints to existing REST APIs
- Improved internal C++ interfaces and data structures
- Fixed bugs on existing products
- Added new interfaces using Unix domain socket for communication with different client/services

Software Engineer *Universidad de Pinar del Rio*

Pinar del Rio, Cuba 01/2018 - 08/2020

- Designed and implemented applications to collect and visualize vibration signals using C++ and Qt
 - Real-time signal processing
 - Technologies used include: C++ — IMU — Arduino — Serial communication — Qt
- Designed and implemented applications to collect and visualize data from sensors for object position tracking
 - Technologies used include: C++ — IMU — Arduino — Serial communication — Linux — OpenGL
- Assisted in teaching programming courses
 - Prepared lectures on C++, Python, Arduino, Linux, Raspberry Pi

- Conducted lab sessions and provided assistance to students with programming projects
- Designed and graded assignments
- ICPC Judge for the Caribbean region
 - Evaluated and proposed challenging programming problems and solution for the ICPC Caribbean Finals
- ICPC coach
 - Coached a team to the ICPC Latin American Finals

DevOps Engineer [Universidad de Pinar del Rio](#)

Pinar del Rio, Cuba 09/2017 - 12/2017

- Setup up multiple servers

Education

B.Sc Telecommunications and Electronics Engineering [UPR](#)

Pinar del Rio, Cuba 09/2012-07/2017

- GPA 4.95/5.00
- ICPC world finalist

Achievements

- **Honourable Mention** 2018 ACM ICPC World Finals
- **2nd Place (Silver Medal)** 2016 ACM ICPC Latin American Finals
- **4th Place** 2015 ACM ICPC Latin American Finals
- **Top 1000** 2019 Google Code Jam

Projects

- **Baseball Limits 2D** ([live demo](#))
 - What does “best ever” look like when you compare two baseball stats at once? In one dimension it’s a ranking; in two it’s a Pareto frontier — the players no one strictly outperforms in both
 - Interactive viewer over 150 years of MLB data (Lahman Database 1871–2023, Baseball Reference 2024)
- **C++ project template**
 - Docker environment with common tools
 - CMake presets for multiple configurations
 - CI pipelines for reporting code coverage and generating documentation
- **DNS library**
 - Parse and generate DNS messages
 - Provide examples of DNS resolution
- **Suffix automaton visualization**
 - Interactive visualization tool using TypeScript and D3 to illustrate the algorithm to build a suffix automaton
- **Template metaprogramming examples**
 - C++ examples to create and modify types and constants at compile time
- **Router-network Docker testbed**

- Docker setup that simulates a small home network — clients behind a router-as-gateway talking to remote servers — for testing network applications end-to-end without real hardware
- **Competitive programming archive:** [reference](#) & [solutions](#)
 - Personal notebook of algorithms and data structures, the kind ICPC teams build up over years
 - Running set of solutions to online-judge problems

Skills

- C — C++ — Python — Bash — CMake — GNU Make — Linux — Buildroot — OpenWrt — Git — Docker — LXC — Algorithms — Data Structures — Networking — IP — DHCP — UDP — TCP — DNS — HTTP — TLS — QUIC

Languages

- **Spanish** [Native]
- **English** [Full Professional Proficiency]